

# Deployment App

Project Brief — 8 Week Intern Project

## Project Overview

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Build a Steam-style deployment system that lets authorized users download and install digital twin deployments on their machines. Each deployment is a self-contained package (an Unreal executable folder, two web folders, and a batch script that we provide). Multiple versions of each deployment must be able to coexist on the same machine so users can keep both stable and beta releases.

The system has two products: a web admin panel for managing users and deployments, and a lightweight Windows launcher app that handles downloads and lets users open the install folder.

*The brief below describes **what** the system should do. The team is expected to propose the technology choices and architecture in Week 1 and present them for sign-off.*

## What the System Does

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### Admin Web Panel

- Admin logs in with a username and password.
- Create, edit, disable, and reset passwords for users.
- Create deployments and add new versions to each deployment.
- Mark each version as either **stable** or **beta**.
- Mark each version as **released** (visible to users) or **archived** (hidden).
- Control which users can see which deployments.
- Adding a new version: admin places the deployment folder on the server, then triggers an "Add version" action in the panel.

### Launcher Client

- Simple, clean design. White background. Logo is swappable so we can rebrand it per client.
- User logs in with the same username and password.
- Sees only the deployments they have been given access to.
- Sees every released version of each deployment, grouped by stable and beta.
- Downloads run fast — files come in via multiple parallel streams.

- Downloads are resumable. If the network drops or the machine restarts mid-download, it picks up where it left off.
- Pause, resume, and cancel buttons.
- Progress UI shows speed, remaining size, and time left.
- Verifies that each downloaded file is complete and not corrupted.
- Multiple versions can be installed side-by-side. A user can have stable 1.2 and beta 1.3 installed at the same time.
- User picks the root folder where deployments get installed.
- Each installed version has an "Open folder" button so the user can run the batch file we provide.
- Checks for free disk space before starting a download.
- Friendly error messages when things go wrong.

## Confirmed Decisions

|                       |                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| Login                 | Simple username + password.                                                                                       |
| Versions on a machine | Side-by-side. Stable and beta can be installed together.                                                          |
| Updates               | Users are not forced to update. They can stay on an older version.                                                |
| Adding a version      | Admin drops the folder on the server, then clicks to add it in the panel.                                         |
| Install location      | User picks the install root folder in the launcher settings.                                                      |
| Launching the app     | For the main project, the user opens the install folder and runs the batch file themselves. See bonus objectives. |

## 8-Week Timeline

*The team self-organises within each week. One or two demonstrable milestones per week. Friday demo every week so progress is visible.*

### Week 1 — Design

- Wireframes and user flows for both the admin panel and the launcher, signed off by mid-week.
- Technology choices proposed and agreed.

### Week 2 — Foundations

- Login working on both the admin panel and the launcher.
- First proof-of-concept download working end-to-end.

### Week 3 — Admin Core

- User management fully working in the admin panel.
- Adding deployments and versions working: admin can drop a folder on the server and register it via the panel.

### Week 4 — Admin Complete, Launcher Connects

- Admin can fully manage users, deployments, versions, and access.
- Launcher logs in and shows each user the deployments they have access to.

### Week 5 — Download Manager

- Full download functionality: parallel streams, resumable, pause / resume / cancel, file integrity verified.

### Week 6 — Integration + Branding, Feature Freeze

- Side-by-side version installs, user-configurable install folder, swappable logo, "Open folder" button, disk space checks, friendly error messages.
- End-to-end demo with a real multi-GB Unreal build on a clean machine.

### Week 7 — Hardening + Debug

- Stress tested: slow network, dropped network mid-download, corrupted files, fresh machine.
- All Week 6 demo bugs fixed.

### Week 8 — Documentation + Handover

- Operations guide (how to publish a new version), admin user guide, handover document.
- Final demo and sign-off.

## Bonus Objectives

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*If the core scope is on track by the end of Week 6, pick any of the following. None are required for sign-off.*

- **Launch deployments from the app.** Run the deployment from inside the launcher, show a "Running" state, and have a "Stop" button that cleanly closes everything that was launched.
- **Auto-install prerequisites on first run.** Detect whether the required software is already installed and offer to install it if missing.
- **Launcher auto-update.** The launcher checks for a newer version of itself on startup and updates in place.

- **Bandwidth cap setting.** User can limit download speed so it doesn't saturate the office network.
- **Port conflict detection.** Before launch, check that the ports the deployment needs are free; show a clear error if not.
- **Download history in admin.** Show who downloaded what and when, with simple filters.
- **Automatic error reporting.** When something goes wrong, the launcher uploads its logs so we can debug.

## Final Deliverables

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- A working admin web panel.
- A working launcher installer.
- Operations guide (how to publish a new version).
- Admin user guide.
- Handover document covering how everything fits together and how to deploy it.